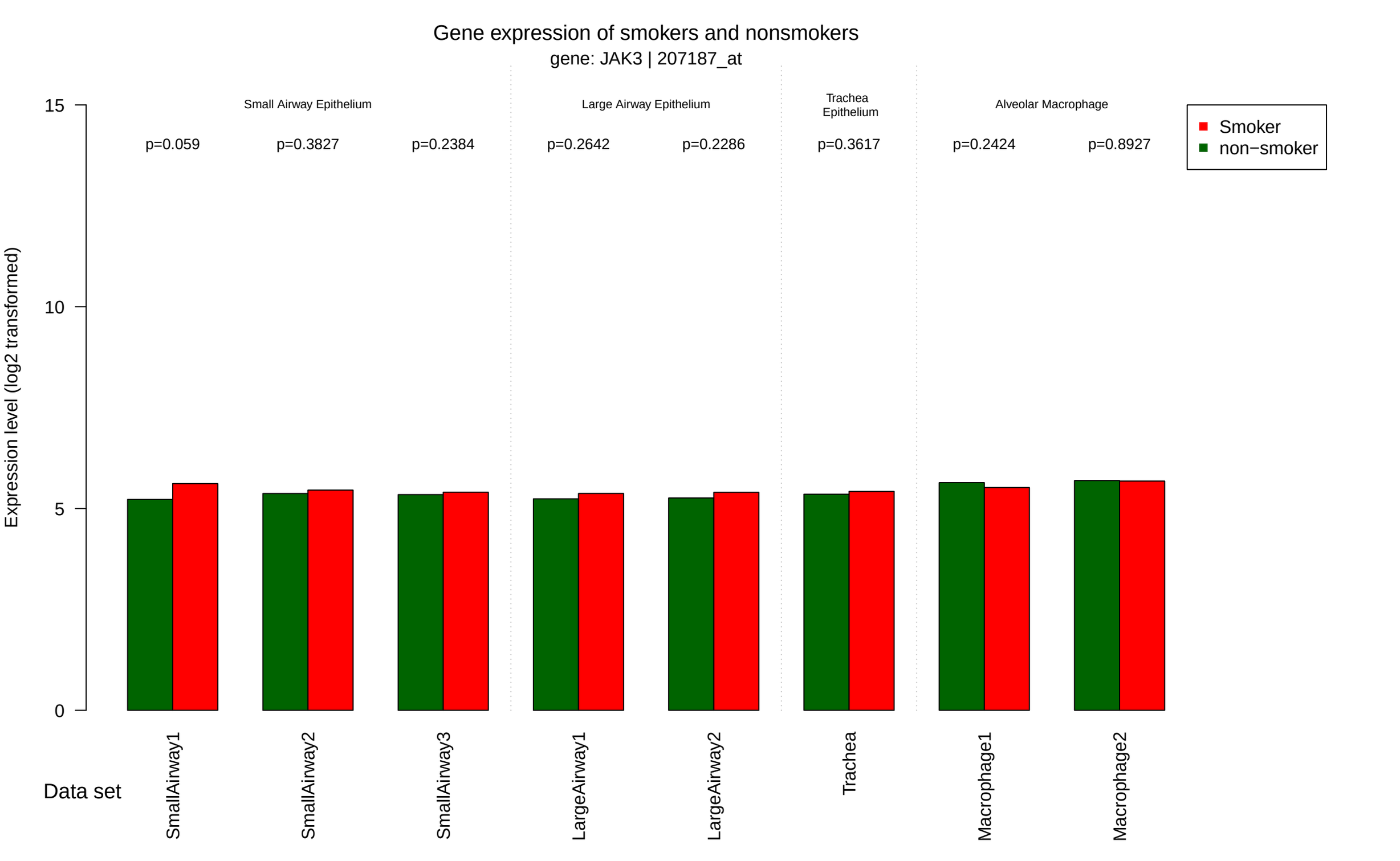
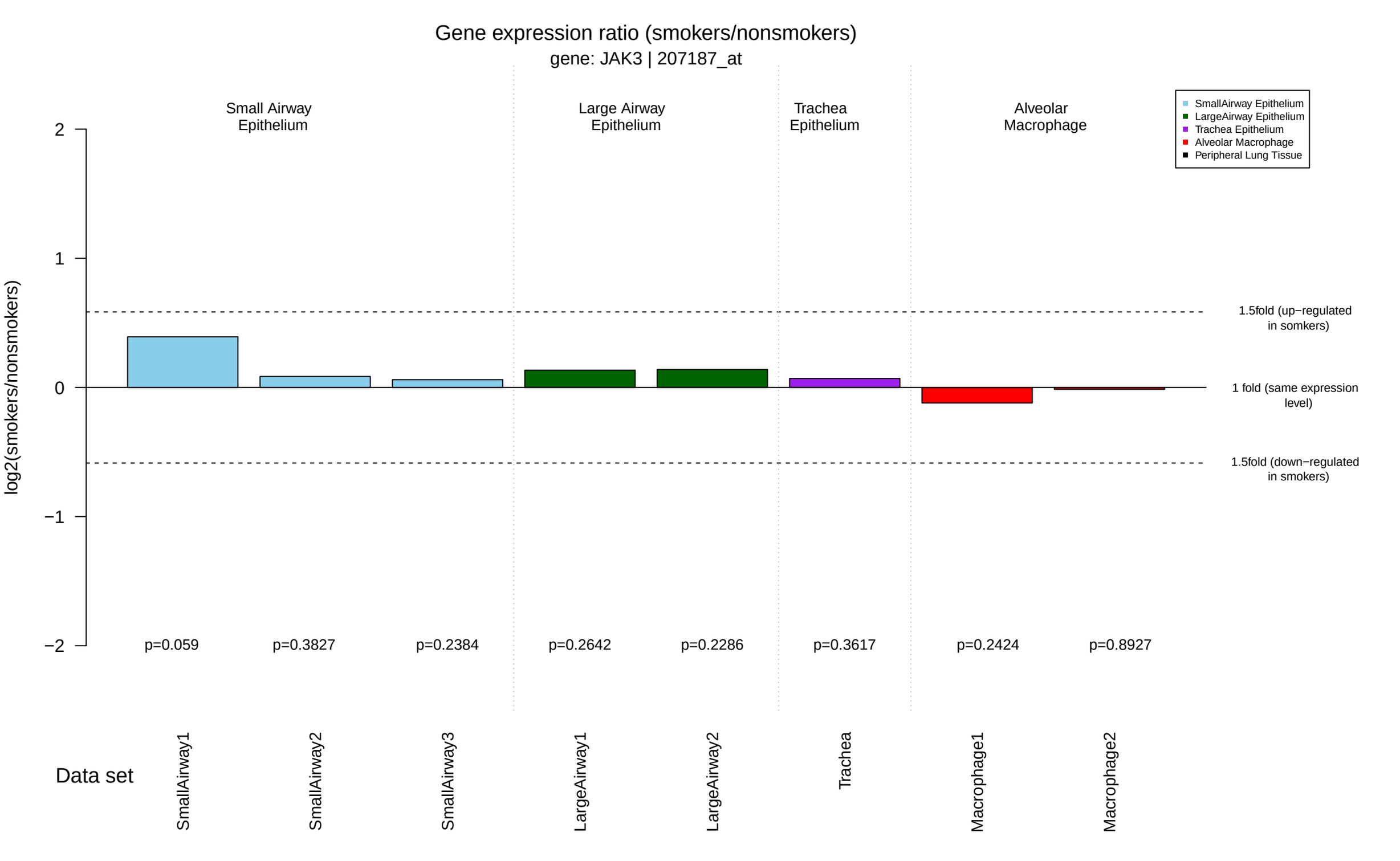
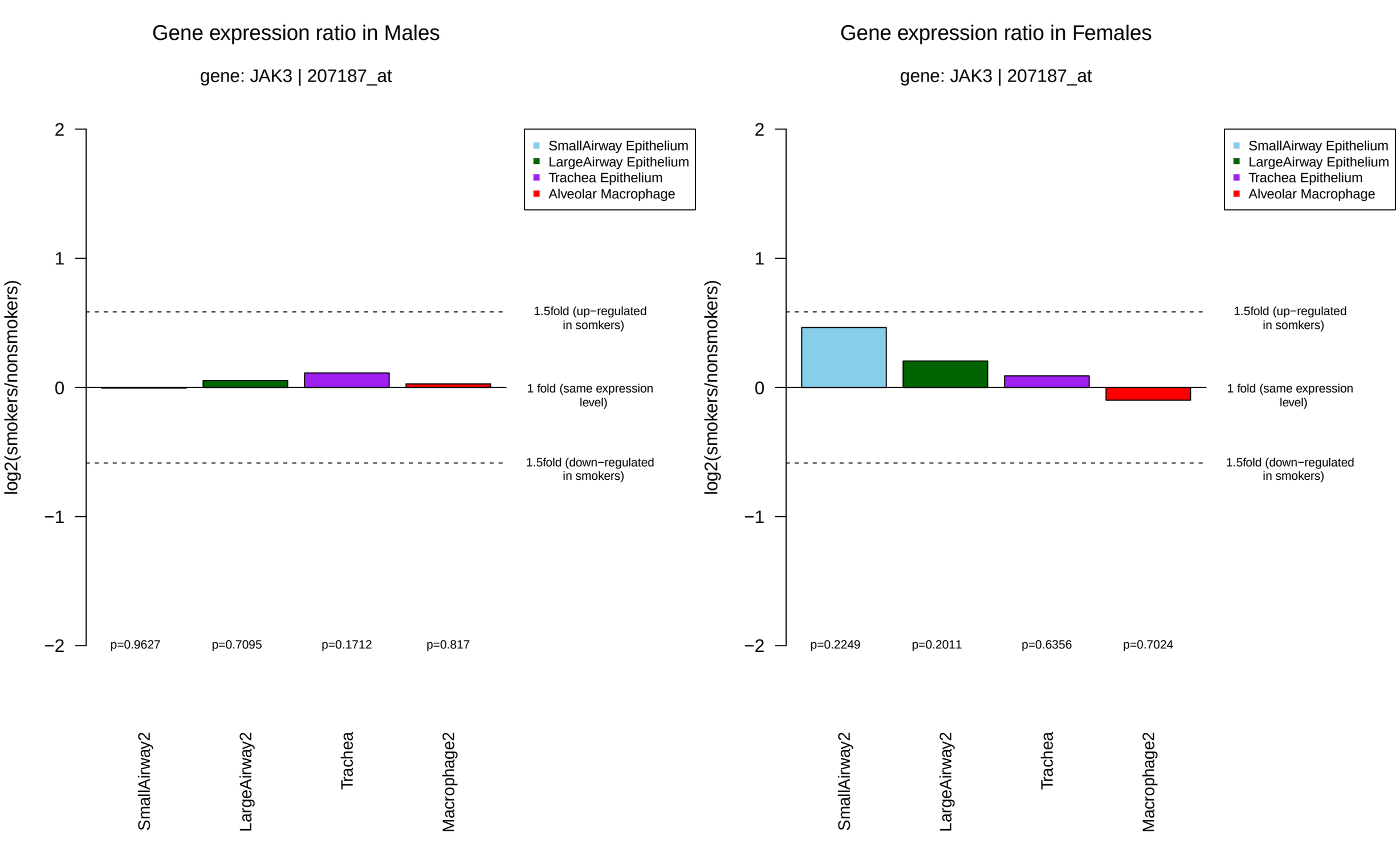
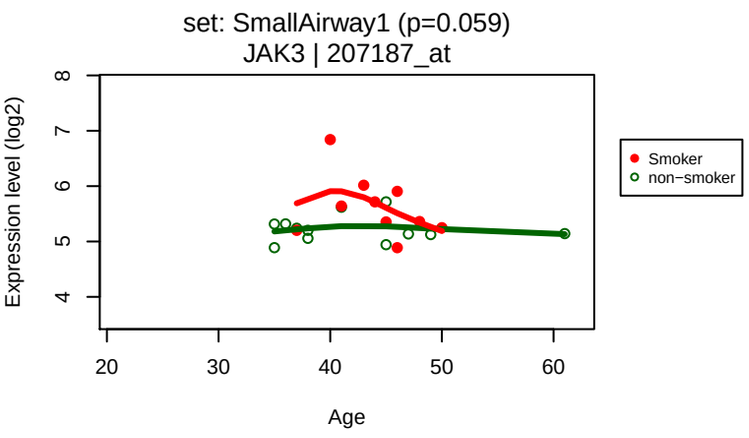
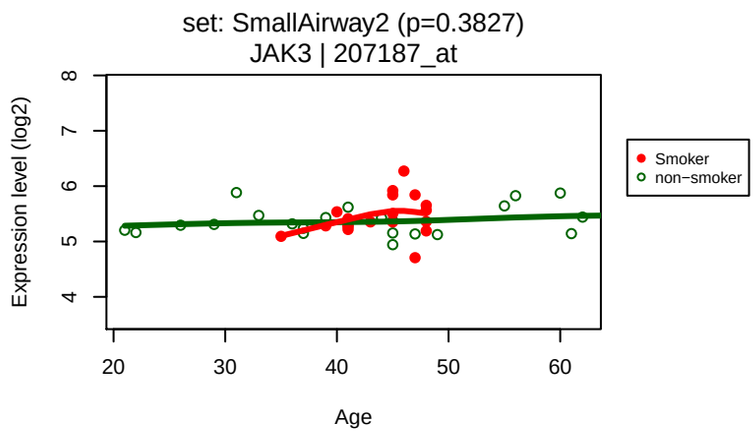
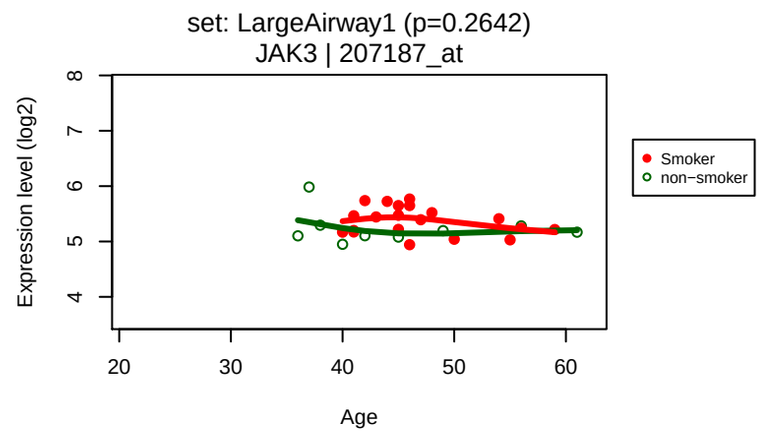

Gene expression of smokers and nonsmokers
gene: JAK3 | 207187_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.059
p=0.3827
p=0.2384
p=0.2642
p=0.2286
p=0.3617
p=0.2424
p=0.8927
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

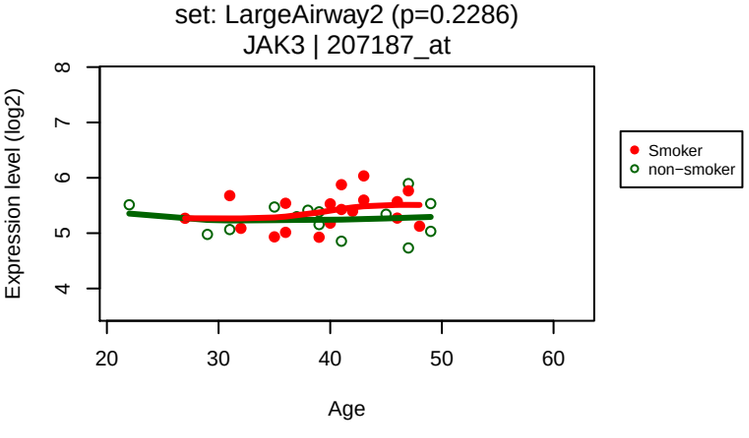

Gene expression ratio (smokers/nonsmokers)
gene: JAK3 | 207187_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.059
p=0.3827
p=0.2384
p=0.2642
p=0.2286
p=0.3617
p=0.2424
p=0.8927
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

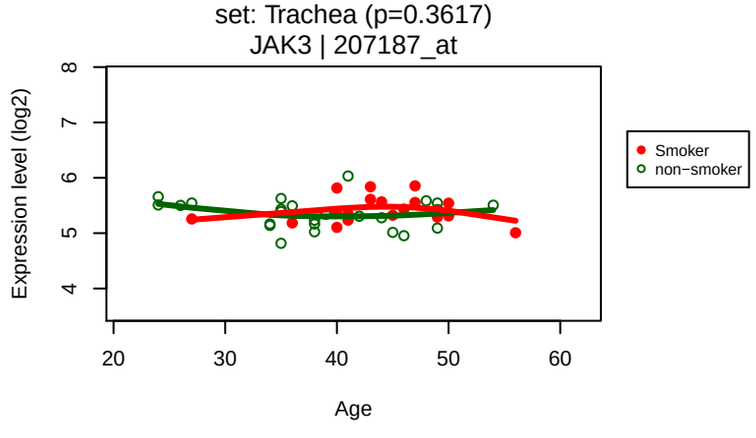

Gene expression ratio in Males
gene: JAK3 | 207187_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9627
p=0.7095
p=0.1712
p=0.817
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: JAK3 | 207187_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2249
p=0.2011
p=0.6356
p=0.7024
SmallAirway2
LargeAirway2
Trachea
Macrophage2

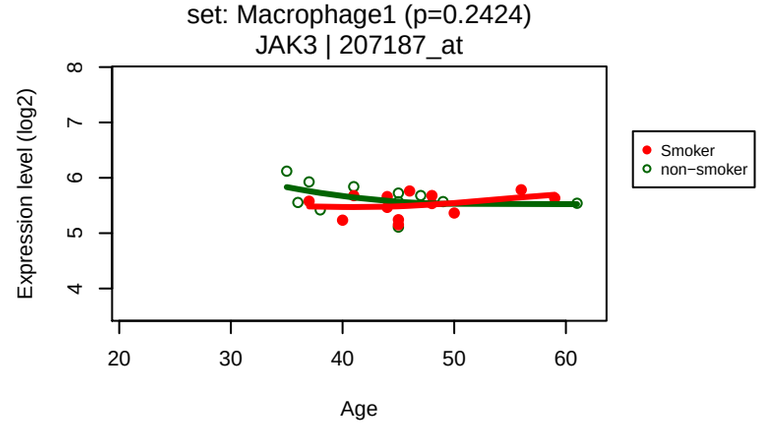

set: SmallAirway1 (p=0.059)
JAK3 | 207187_at
Expression level (log2)
Age
Smoker
non−smoker

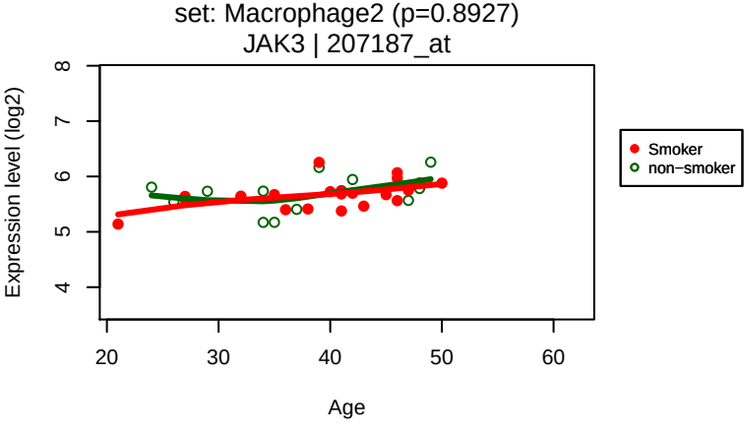

set: SmallAirway2 (p=0.3827)
JAK3 | 207187_at
Expression level (log2)
Age
Smoker
non−smoker

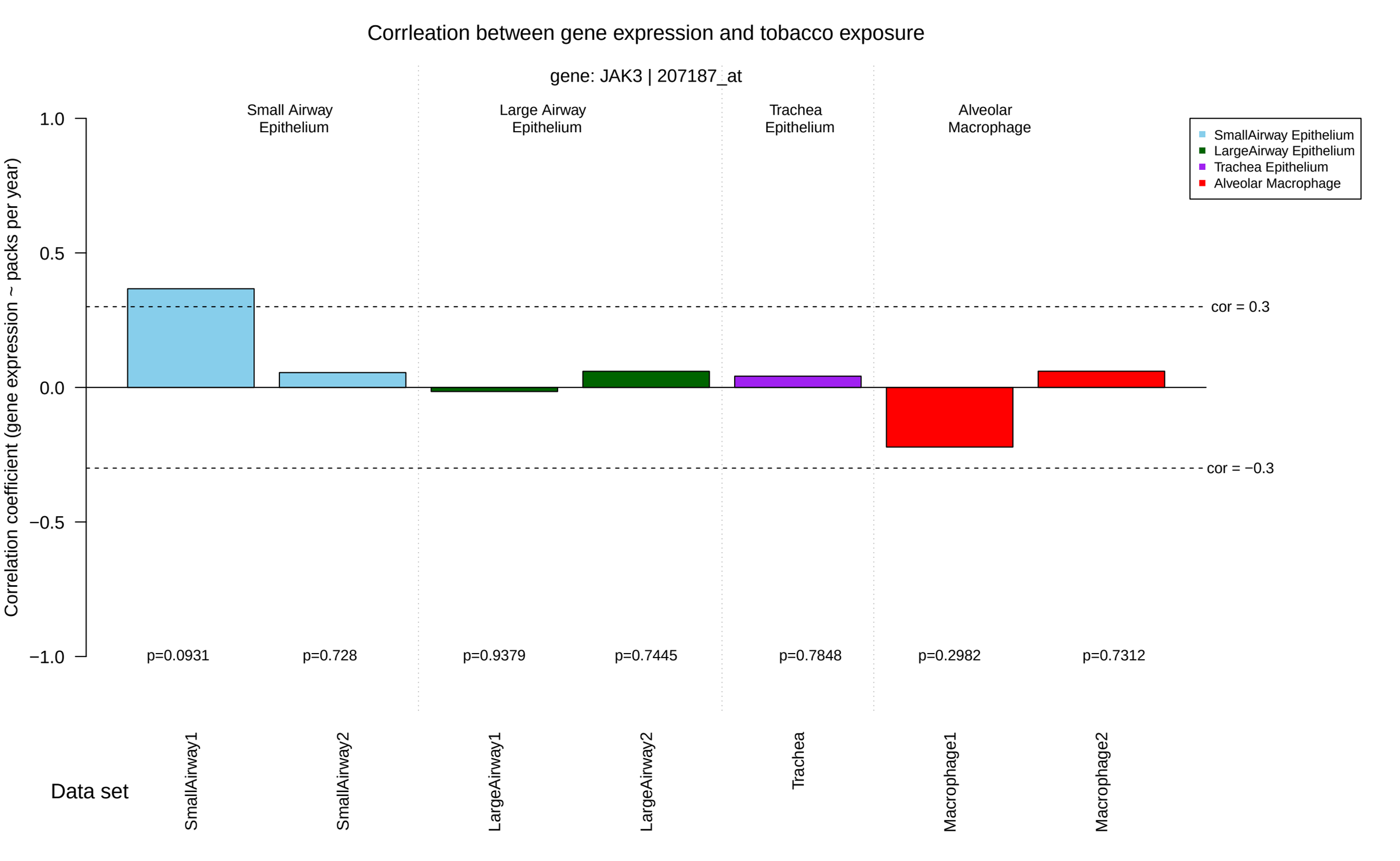

set: LargeAirway1 (p=0.2642)
JAK3 | 207187_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2286)
JAK3 | 207187_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3617)
JAK3 | 207187_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2424)
JAK3 | 207187_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8927)
JAK3 | 207187_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: JAK3 | 207187_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0931
p=0.728
p=0.9379
p=0.7445
p=0.7848
p=0.2982
p=0.7312
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2